

🔥 XAI: Quantifying Data Value and Influence in Machine Learning

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer -----

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

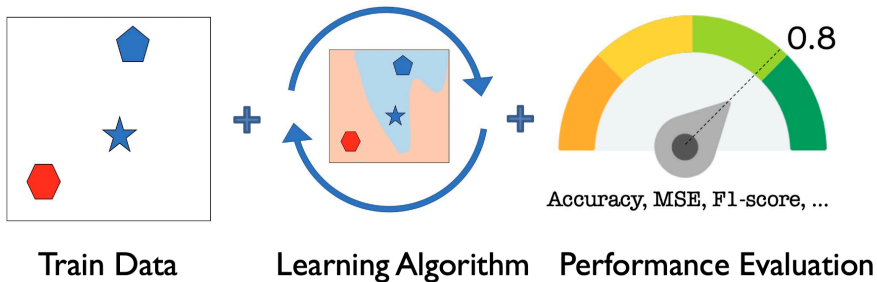
Quantifying Data Value and Influence in Machine Learning

Combining insights from:

What is Your Data Worth? Equitable Valuation of Data (Ghorbani & Zou, ICML 2019)

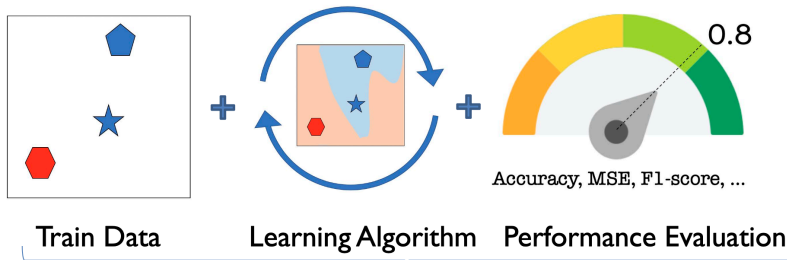
Understanding black-box predictions via influence functions (Koh & Liang, ICML 2017)

🍲 Ingredients of Data Value -----

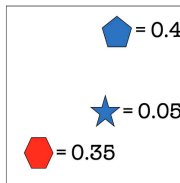


- How can we **split** the 0.8 Performance Value into the Train Data?

🍲 Ingredients of Data Value -----



Value depends
on the learner,
evaluation and
dataset.



Value

There are many
ways to "value"
data.
Is there one
right way?

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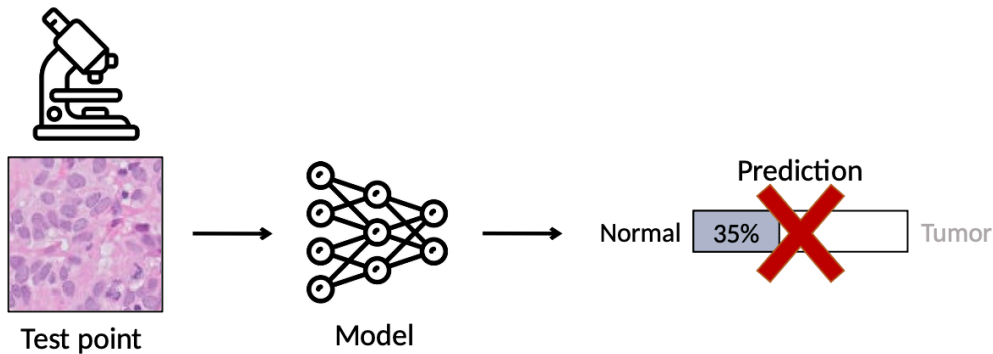
🍷 The Data Valuation & Influence Problem -----

We need a principled way to measure its value and its effect on our models.

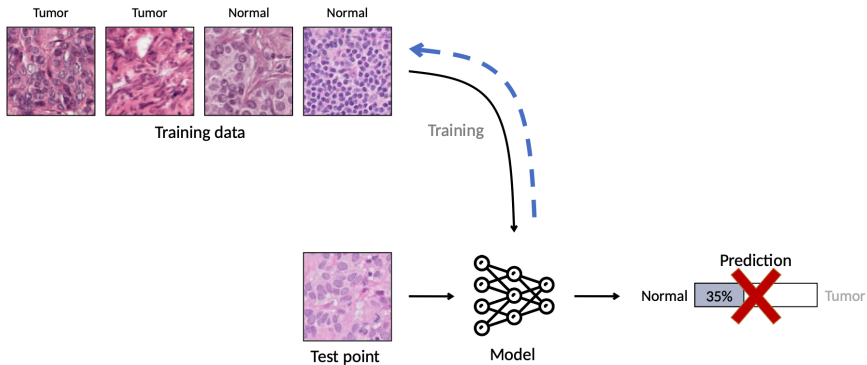
Stakeholders have different priorities:

- **ML Engineers:** Assess heterogeneous sources, debug model behavior, ensure data quality, curate mislabeled data, etc.
- **Data Vendors:** Determine fair pricing for buying and selling data.
- **Individuals/Data Producers:** Understand compensation or credit for contributed data.
- **Regulators & Auditors:** Ensure compliance, fairness, and accountability in data usage.
- **Researchers:** Benchmark datasets, study data efficiency, and advance reproducibility.

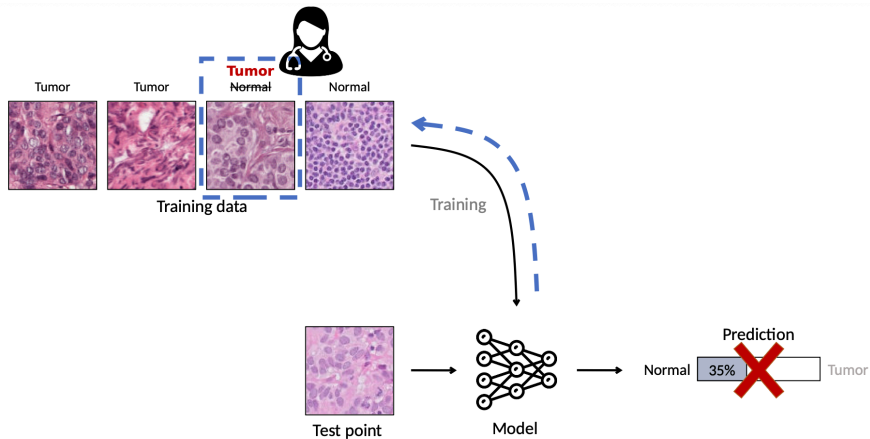
🍷 The Baseline: Leave One Out Method -----



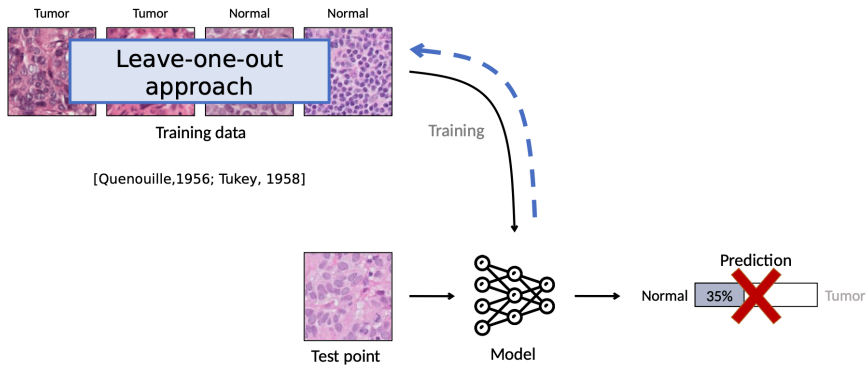
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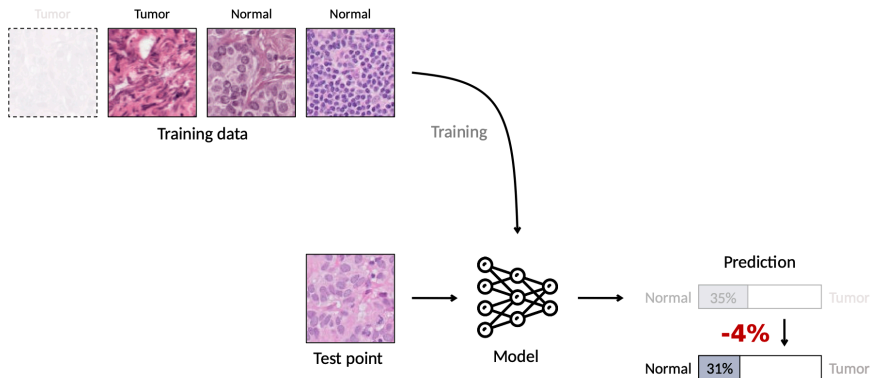
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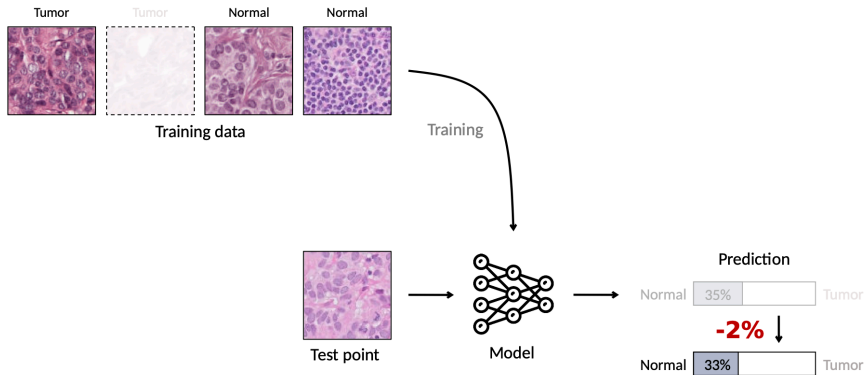
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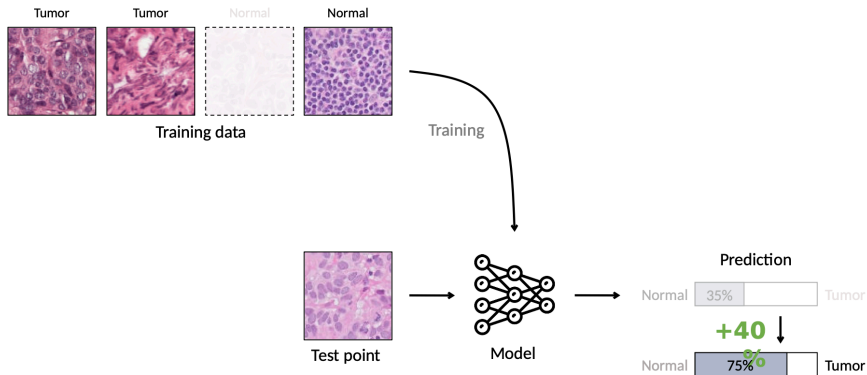
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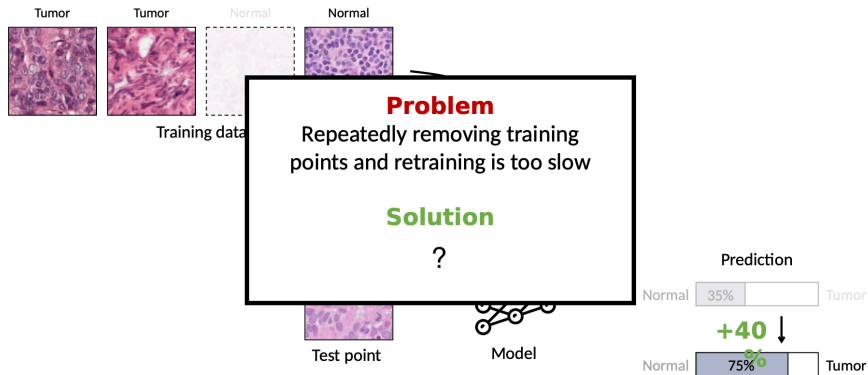
🍵 The Baseline: Leave One Out Method -----



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🍲 The Baseline: Leave One Out Method -----



The Bottleneck -----

Leave-One-Out (LOO) is conceptually simple but practically flawed:

- ① **Computationally intractable:** Retraining the model for every single data point is impossible for large datasets.
- ② **Ignores interactions:** It doesn't account for how data points interact with one another in subsets.

Two frameworks address these issues: **Data Shapley** and **Influence Functions**.

Approach 1: Data Shapley

Equitable Valuation of Data

(Ghorbani and Zou 2019)

🍷 Desirable Properties of Valuation -----

A fair valuation method must satisfy three fundamental axioms from cooperative game theory:

- ① **Null Element:** If adding a point to *any* subset of training data never changes the model's performance, its value is exactly 0.
- ② **Symmetry:** If adding point A or point B to any subset always results in the exact same performance change, they must receive the same value ($Value(A) = Value(B)$).
- ③ **Linearity:** If the performance metric is a sum of performances on individual tasks, the data value must reflect the sum of values for those individual tasks. Add or remove a task should add or remove value for that task.

These are the Shapley axioms.

The Data Shapley Value -----

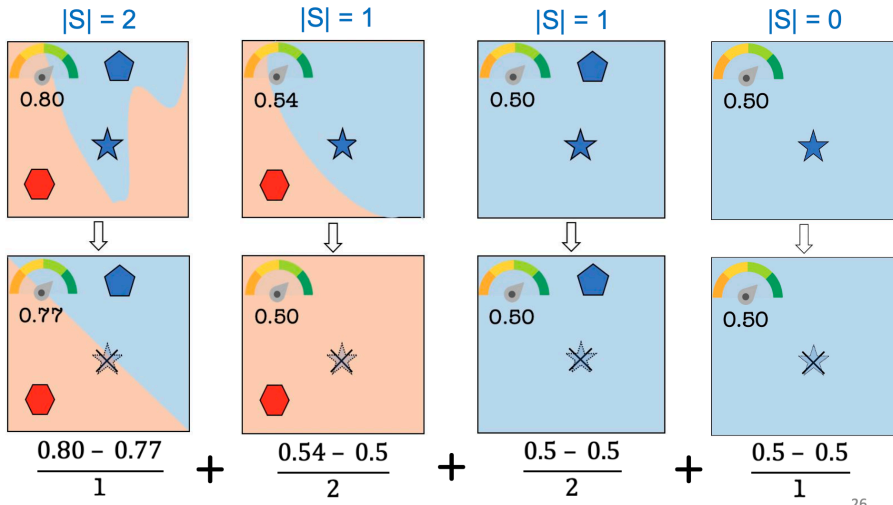
The only data value mathematically guaranteed to satisfy these three properties is based on marginal contributions across all possible subset sizes.

$$Value(data\ k) = \sum_{S \subseteq Dataset \setminus \{k\}} \frac{performance(S \cup \{k\}) - performance(S)}{\binom{n-1}{|S|}}$$

- Evaluates the expected marginal contribution to all possible sizes of train data samples.
- Directly applies Lloyd Shapley's cooperative game theory to individual ML data points.

🥥 The Data Shapley Value Example -----

Example: value (★) = 0.05



🍷 Efficient Approximation: TMC-Shapley -----

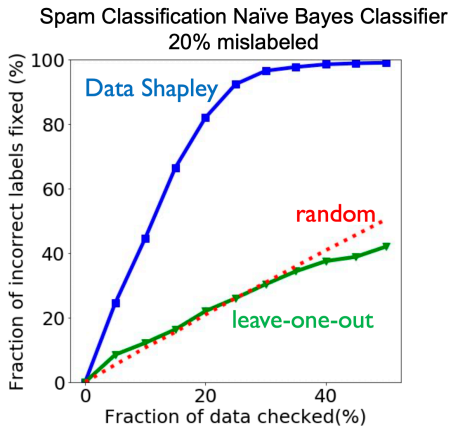
Exact calculation is computationally infeasible for realistic datasets because it grows exponentially.

Truncated Monte-Carlo (TMC) Approximation:

- ➊ Sample a random permutation of the training data.
- ➋ Add one point at a time based on the sampled permutation.
- ➌ Re-train the model to capture the marginal increase in performance.
- ➍ **Truncate** the calculation when performance saturates to save computational resources.

🍲 Application 1: Identifying Low-Quality Data -----

Result: Ordering points from most negative Shapley value to positive allows for rapid dataset cleaning. Examining just ~30% of the data uncovers nearly all mislabeled examples.



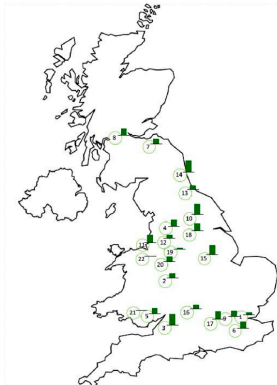
🍲 Application 2: Identifying Essential Data -----

Case Study: UK Biobank

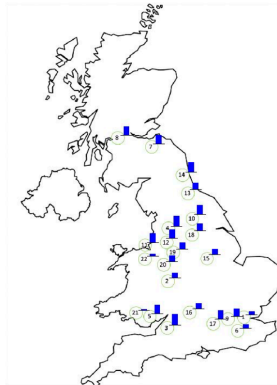
- Dataset spanning 500,000 individuals across 22 centers in the UK.
- Centers were evaluated as singular “data sources” for predicting Breast and Colon Cancer.

🍷 Application 2: Identifying Essential Data -----

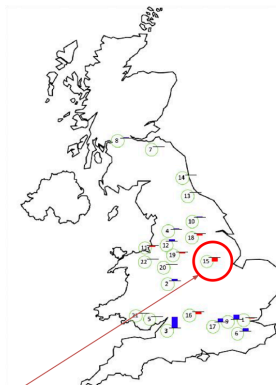
of patients



Breast Cancer



Colon Cancer



Most predictive feature: Age $\uparrow$ $\square$ Colon Cancer $\uparrow$ ($p=1.5e-6$)
Center-14: cancer unrelated to age ($p=0.14$)

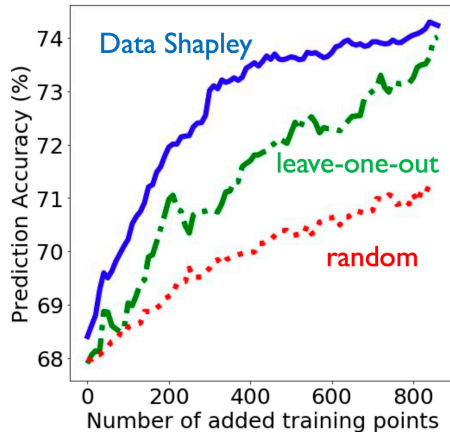
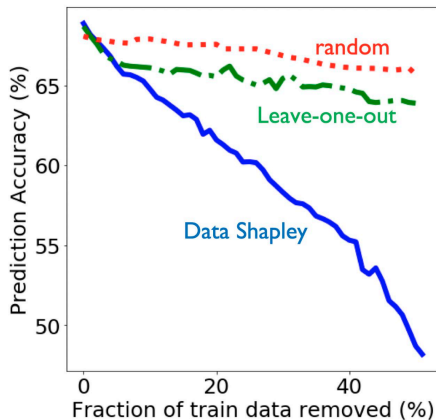
🍲 Application 2: Identifying Essential Data -----

Insights from Shapley:

- For most centers, its value is related to the number of patients. This is consistent for Breast Cancer.
- For Colon Cancer, a massive center received a *negative* Shapley value.
- Why this specific center? We don't know for sure.
- A hypothesis is that the model heavily relies on *Age* as a predictive feature, but the specific anomalous center exhibited colon cancer rates entirely independent of age. We detected out-of-distribution data.



Application 2: Identifying Essential Data -----



Application 3: Domain Adaptation -----

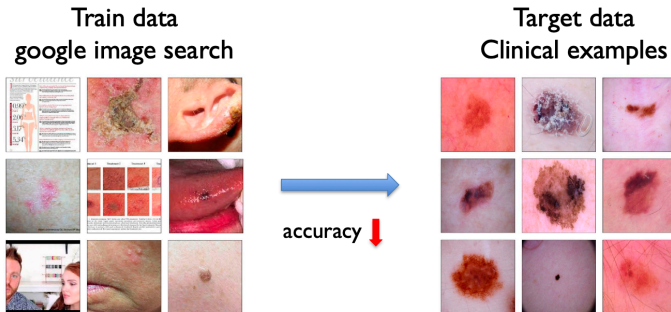
What if the training data differs in quality, distribution, or class balance from the test environment?

Shapley Adaptation Strategy:

- 1 Compute Shapley values relative to the target test distribution.
- 2 Remove data with negative values.
- 3 Reweight the remaining training data based on relative positive weights.

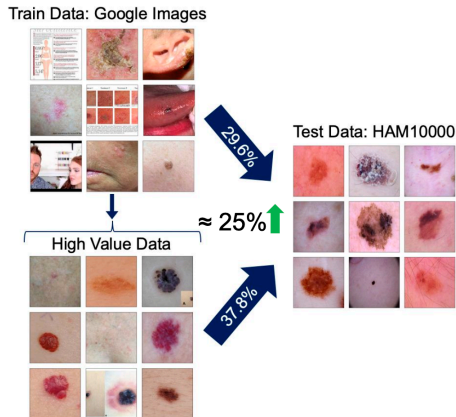
🍲 Domain Adaptation Results (Skin Lesion) -----

Training on noisy Google Image Search data to test on the clean, clinical HAM10000 dataset.



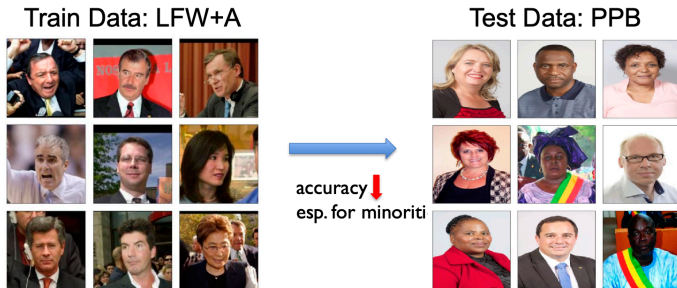
🍵 Domain Adaptation Results (Skin Lesion) -----

Applying Shapley weights improved classification by $\approx 25\%$.



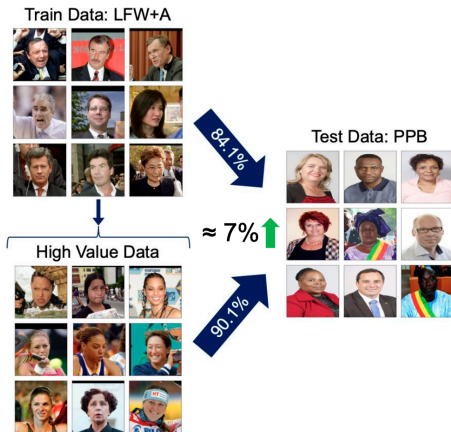
🍲 Domain Adaptation Results (Gender Detection) -----

Models trained on LFW+A (heavily biased toward white males) fail on balanced datasets like PPB.



🍲 Domain Adaptation Results (Gender Detection) -----

Shapley values identify out-of-distribution elements; adaptation increases accuracy on underrepresented minorities by $\approx 7\%$. Top 20% of high-value datapoints are all female.



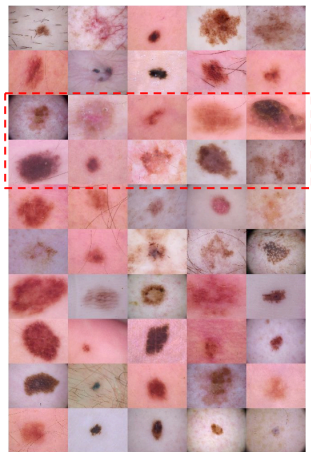
Extra: Distributional Shapley

A More Intrinsic Measure of Value

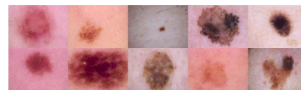
(Ghorbani et al. 2020)

🍲 Extra! (Distributional Shapley) -----

Seller



Buyer

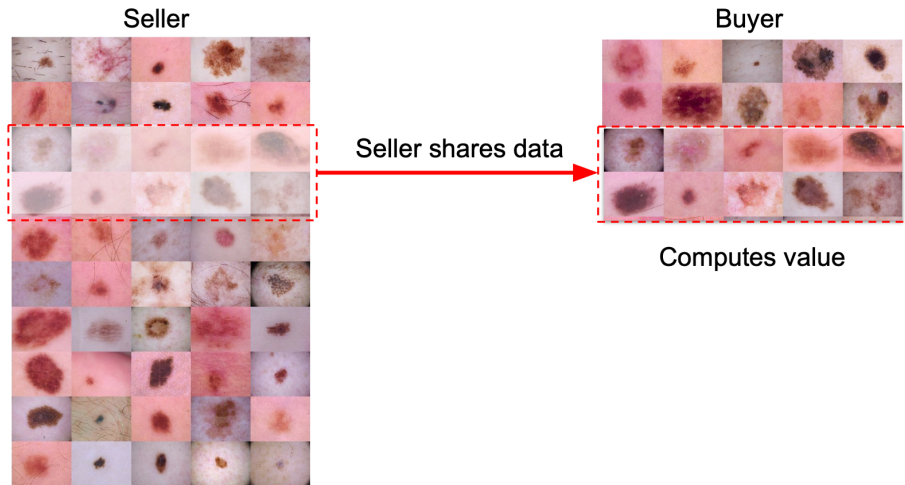


🍷 Extra! (Distributional Shapley) -----

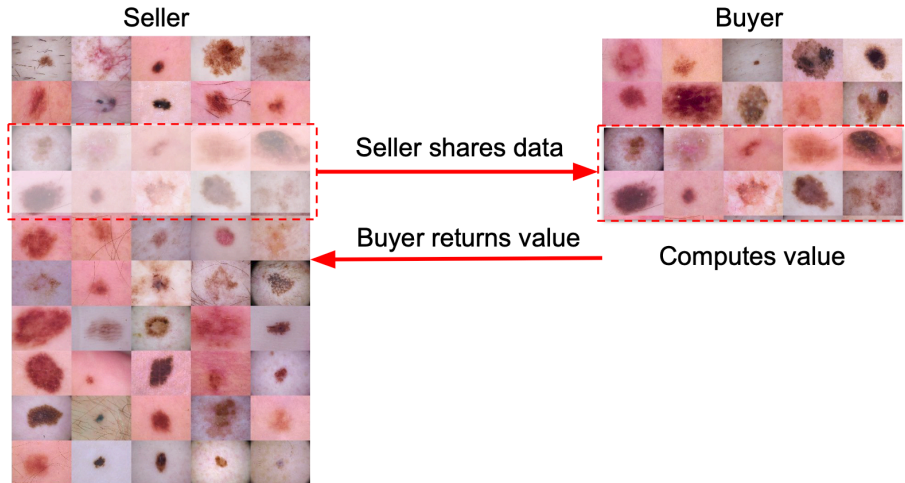


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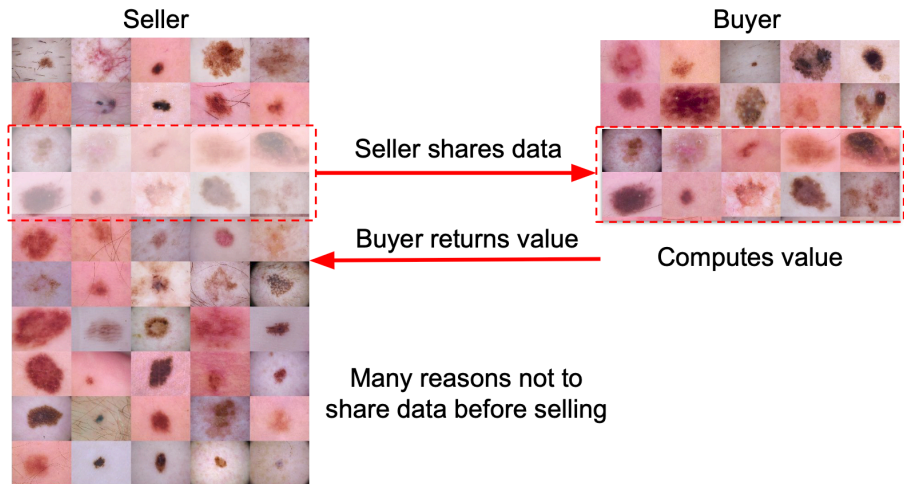
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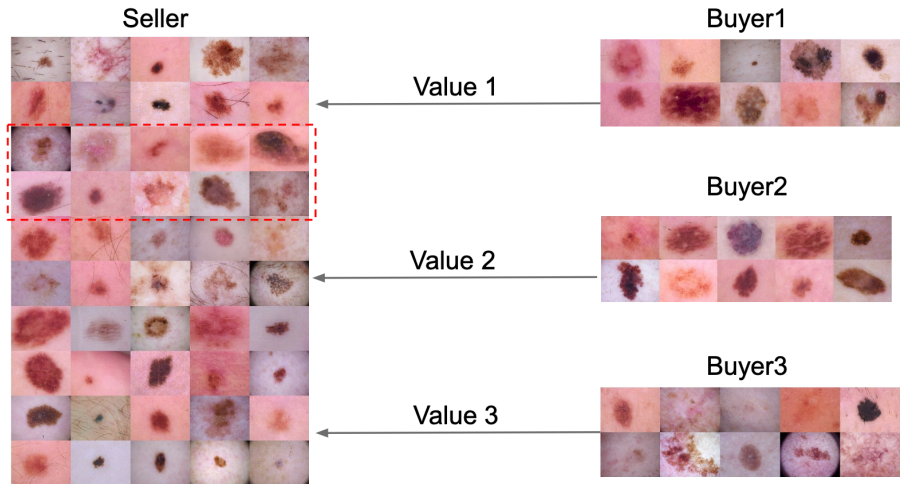
🍷 Extra! (Distributional Shapley) -----



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🍷 Extra! (Distributional Shapley) -----



Approach 2: Influence Functions

Understanding black-box predictions via influence functions

(Koh and Liang 2017)

🍷 The Influence Function Approximation -----

Change in loss on after removing

Model's representation of

Effect of other training points

Model's representation of

$$\ell(z_{test}, \hat{\theta}_{-z_{train}}) - \ell(z_{test}, \hat{\theta}) \approx \nabla_{\theta} \ell(z_{test}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} \ell(z_{train}, \hat{\theta})$$

Gradient of loss on

Inverse of the Hessian

Gradient of loss on

The diagram illustrates the Influence Function Approximation equation. The equation is: $\ell(z_{test}, \hat{\theta}_{-z_{train}}) - \ell(z_{test}, \hat{\theta}) \approx \nabla_{\theta} \ell(z_{test}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} \ell(z_{train}, \hat{\theta})$. Annotations include: a bracket above the left side labeled 'Change in loss on after removing'; a bracket above the first term of the right side labeled 'Model's representation of' with 'Gradient of loss on' below it; a bracket above the second term of the right side labeled 'Model's representation of' with 'Gradient of loss on' below it; a vertical arrow pointing down to the $H_{\hat{\theta}}^{-1}$ term labeled 'Effect of other training points'; and a vertical arrow pointing up to the $H_{\hat{\theta}}^{-1}$ term labeled 'Inverse of the Hessian'.

Doesn't require retraining!

Interpretation: Model Representations -----

$$\ell(z_{\text{test}}, \hat{\theta}_{-z_{\text{train}}}) - \ell(z_{\text{test}}, \hat{\theta}) \approx \nabla_{\theta} \ell(z_{\text{test}}, \hat{\theta})^T H_{\hat{\theta}}^{-1} \nabla_{\theta} \ell(z_{\text{train}}, \hat{\theta})$$

Gradient = model's representation

$$\nabla_{\theta} \ell(z_{\text{test}}, \hat{\theta})$$

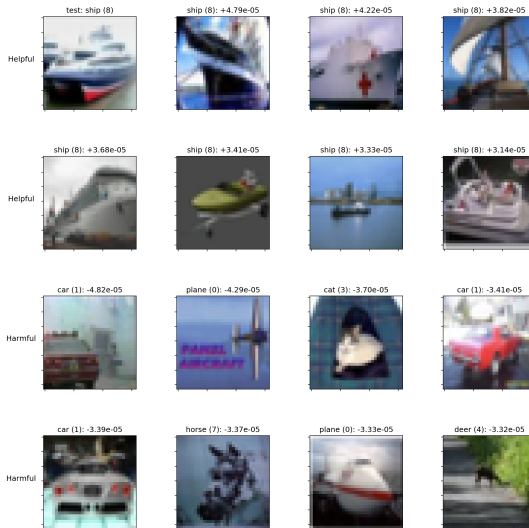
How "loudly" the training point wants to pull the model parameters to reduce its own error.

$H_{\hat{\theta}}^{-1}$ = curvature of overall loss

A measure of the curvature of the loss function across all training data. It essentially calculates how strongly the rest of the training dataset resists the parameter changes pushed by point z .

Implementation Output -----

Training sample influences on test outcome of resnet_basic_110



🍷 Limitations -----

Key assumptions required

- **Convexity:** derivation assumes a convex loss landscape
- **Convergence:** derivation assumes the optimizer has converged

Can you calculate influence functions for a neural network?

- Neural networks are inherently **non-convex**.
- Frequently feature **non-differentiable** components.
- Parameters are typically obtained via **SGD with early stopping**, rather than converging to a true global minimum.

⇒ To adapt influence functions to neural networks, the paper introduces several **practical approximations**.

Summary -----

Both approaches aim to link model behavior back to the training data.

Data Shapley:

- Axiom-backed framework moving beyond LOO by measuring expected contributions over all subsets.
- Excellent for data valuation, pricing, cleaning, and domain adaptation.
- Computationally intensive (requires Monte Carlo approximation).

Influence Functions:

- Efficient gradient-based calculation—no retraining needed.
- Excellent for fast debugging, fairness analysis, and understanding specific predictions.
- Limited by strict mathematical assumptions (convexity, convergence) in complex models.

Thank You!

References ----- |

- Ghorbani, Amirata, Michael P. Kim, and James Zou. 2020. "A Distributional Framework for Data Valuation." *Proceedings of the 37th International Conference on Machine Learning*, ICML'20.
- Ghorbani, Amirata, and James Zou. 2019. "Data Shapley: Equitable Valuation of Data for Machine Learning." In *Proceedings of the 36th International Conference on Machine Learning*, edited by Kamalika Chaudhuri and Ruslan Salakhutdinov, vol. 97. Proceedings of Machine Learning Research. PMLR.
<https://proceedings.mlr.press/v97/ghorbani19c.html>.
- Koh, Pang Wei, and Percy Liang. 2017. "Understanding Black-Box Predictions via Influence Functions." *International Conference on Machine Learning*, 1885–94.